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Method of manufacturing press-formed product, and tray and hot-press manufacturing line used for manufacturing press-formed product

Abstract

A method of manufacturing a press-formed product includes a heating step, a transportation step, and a pressing step. At the heating step, workpieces are heated while a first workpiece W1 is placed on a first group of struts 3 extending upward from a tray body 2, and a second workpiece W2 is placed on a second group of struts 3 to be located above the first workpiece W1 and overlap it. At the transportation step, the second workpiece W2, while located above the first workpiece W1 and overlapping it, is transported, together with the tray body 2, from the heating device 14 to the lifting location. At the lifting location, the second workpiece W2 is lifted upward by the transportation device 46 and the first workpiece W1 is lifted by the transportation device 46 before they are transported to their respective pressing locations.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a method of manufacturing a press-formed product including the steps of heating, transportation and pressing, and a tray and a hot-press manufacturing line used for manufacturing a press-formed product.

BACKGROUND ART

(2) Techniques are in use that heat material to a predetermined temperature and press the material using a press machine. For example, during hot pressing, material, i.e., steel sheet for hot pressing, is heated to the austenite range (i.e., about 900° C. and higher), and is hot press-formed. Thus, quenching is performed together with the forming process, thereby providing a press-formed product with a strength of the order of 1500 MPa or higher, for example. During typical hot stamping, quenching is performed as the sheet is rapidly cooled by contact heat transfer to the die during press-forming. This means that, to sufficiently produce the effects of quenching, the temperature of the material at the beginning of press-forming, which roughly corresponds to the quenching initiation temperature, must not be lower than a certain temperature. In such cases, this certain temperature at the beginning of press-forming is, for example, 700° C. or higher depending on the material.

(3) Japanese Patent Nos. 5910305 and 5910306 each disclose a hot press-forming method involving positioning a plurality of electrically conductive sheet-shaped workpieces so as to overlap each other, attaching electrodes to the workpieces, and passing electricity through the workpieces to heat the sheet-shaped workpieces. The heated sheet-shaped workpieces are positioned at predetermined press locations that are different from the location where they receive electricity. Each of the sheet-shaped workpieces at the press locations is press-formed. Heating a plurality of sheet-shaped workpieces by simultaneously passing electricity therethrough improves productivity.

(4) In the case of the above-described conventional techniques, heat is diffused out of the workpiece during transportation thereof, causing the temperature of the workpiece to decrease. As a result, when the workpiece is being transported into the die of the press machine, the workpiece may not be maintained at a temperature required from the workpiece, potentially leading to insufficient quenching of the press-formed product.

(5) To solve this problem, Japanese Patent No. 5814669 discloses a transportation device for hot pressing, including a manufacturing line for hot pressing where a panel-shaped object is kept in a heated state and transported from one step to another of the line. The transportation device for hot pressing transports the heated object while the object is covered with a temperature-retaining cover. This maintains the object being transported at a temperature required for quenching.

(6) Japanese Patent No. 4673656 discloses a hot press-forming apparatus including a device for heating a metal sheet that is being processed, including a primary heating means using dielectric heating and electrical heating, and a secondary heating means using radiant heat transmission. The secondary heating means using radiant heat transmission is provided on a device for transportation from the primary heating means to the hot press-forming die. Secondary heating using radiant heat transmission achieves uniform heating of the metal sheet, reducing temperature deviation in the metal sheet.

PRIOR ART DOCUMENTS

Patent Documents

(7) Patent Document 1: Japanese Patent No. 5910305 Patent Document 2: Japanese Patent No.

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

(8) The inventors noted that, if the material to be hot pressed has a reduced thickness, a temperature drop after heating of the material and during transportation to the press machine may affect the quality of the resulting press-formed product. The inventors investigated how to reduce temperature drop in the material being transported. In the course of the investigation, they found that merely covering the material being transported with a temperature-retaining cover, as with the above-described conventional techniques, may not be enough to sufficiently reduce temperature drop. A secondary heating means may be provided to heat the material during transportation. However, this requires additional equipment including a heat source for the secondary heating means provided on the transportation path. This increases the size of the entire equipment and may, at the same time, lead to increased equipment and/or operation costs.

(9) In view of this, the present application discloses a method of manufacturing a press-formed product, as well as a tray, where, in hot press-forming, temperature drop in material during the transportation period after heating of the material until beginning of press-forming is reduced in a simple manner.

Means for Solving the Problems

(10) A method of manufacturing a press-formed product according to an embodiment of the present invention includes: a heating step in which a sheet-shaped first workpiece and a sheet-shaped second workpiece are simultaneously heated; a transportation step in which the first and second workpieces heated at the heating step are transported to a press machine; and a pressing step in which the first and second workpieces transported to the press machine are processed by the press machine. At the heating step, the first and second workpieces are heated by a heating device while the first workpiece is placed on a first group of at least three struts extending upward from a tray body having an empty space extending therethrough in a top-bottom direction when viewed from above and the second workpiece is placed on a second group of at least three struts extending upward from the tray body and positioned above the first workpiece to overlap the first workpiece in a direction normal to a sheet surface of the first workpiece.

(11) The transportation step includes a first transportation sub-step and a second transportation sub-step. At the first transportation sub-step, the first and second workpieces, together with the tray body, are transported from the heating device to a lifting location while the first workpiece is placed on the first group of struts and the second workpiece is placed on the second group of struts and positioned above the first workpiece to overlap the first workpiece in the direction normal to the sheet surface of the first workpiece. At the second transportation sub-step, the second workpiece placed on the second group of struts, when at the lifting location, is lifted upward by a transportation device and transported to a press location for the second workpiece, and the first workpiece placed on the first group of struts, when at the lifting location, is lifted upward by the transportation device and transported to a press location for the first workpiece.

Effects of the Invention

(12) According to the present disclosure, in hot press-forming, temperature drop in material during the transportation period after heating of the material until beginning of press-forming is reduced in a simple manner.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic view of a hot-press manufacturing line according to an embodiment.

- (2) FIG. 2 is a top view of a tray according to an embodiment.
- (3) FIG. 3 is a side view of the tray shown in FIG. 2 as viewed in the direction of arrow F.
- (4) FIG. 4 is a top view of a tray with another exemplary arrangement of the first and second groups of struts.
- (5) FIG. 5 is a side view of the tray shown in FIG. 4 as viewed in the direction of arrow F.
- (6) FIG. 6 is a side view of a tray with groups of struts accommodating first and second workpieces that are not flat sheets.
- (7) FIG. 7 is a side view of a tray with groups of struts accommodating first and second workpieces that are not flat sheets.
- (8) FIG. 8 is a top view of a tray accommodating first and second workpieces W1 and W2 with different surface areas as viewed from above.
- (9) FIG. 9 is a side view of the tray shown in FIG. 8 as viewed in the direction of arrow F.
- (10) FIG. 10 is a side view of a tray accommodating first and second workpieces that are differential-thickness sheets.
- (11) FIG. 11 is a top view of a tray arrangement including shields.
- (12) FIG. 12 is a side view of the tray shown in FIG. 11 as viewed in the direction of arrow F.
- (13) FIG. 13 shows a variation of the hot-press manufacturing line of the embodiments.
- (14) FIG. 14A illustrates a construction of the transportation device shown in FIG. 13.
- (15) FIG. 14B shows the transportation device shown in FIG. 14A with its first arms opened.
- (16) FIG. 15 indicates locations at which temperature is measured in the examples.
- (17) FIG. 16 is a graph of measurements of average temperature drop rate.
- (18) FIG. 17 is a graph indicating the range from which the average temperature drop rates were derived.

EMBODIMENTS FOR CARRYING OUT THE INVENTION

(19) In hot press-forming, the temperature at the beginning of press-forming depends on the temperature to which material is heated and on the amount of temperature decrease after heating of the material in the period of time for which the material is transported to the die for press-forming. The temperature to which material is heated depends on metallurgical conditions. The period of time of transportation to the die after heating depends on the construction and/or specifications of the equipment. The amount of temperature decrease during this transportation depends on the heat capacity of the material. For example, in the case of a steel sheet, heat escapes mainly through heat transfer and heat radiation (or simply radiation) from the upper and lower surfaces to the atmosphere. The inventors noticed that the amount of temperature decrease during transportation significantly depends on the sheet thickness of the material. More specifically, as discussed above, as the sheet thickness of material decreases, the amount of temperature decrease increases even for the same transportation time, potentially making it difficult to provide a forming initiation temperature required for quenching. As a result, there may be cases where a part does not provide a strength required for press-forming.

(20) The inventors did research to find a way to reduce temperature drop during transportation without an additional heat source. When a sheet-shaped material (i.e., workpiece) has been heated by a heating device, the workpiece is transported on a transportation path, such as a set of rollers, from the transportation device to a location at which the workpiece is lifted by a transportation device. The workpiece is lifted by the transportation device and placed on a press location in the press machine. The inventors closely looked at the state in which the workpiece is until being moved out of the heating device and lifted by the transportation device.

(21) After extended research, the inventors arrived at positioning a plurality (e.g., two) sheet-shaped workpieces above a tray so that they overlap in the top-bottom direction, heating them in a heating device and, after heating, transporting them together with the tray from the heating device while they remain overlapping in the top-bottom direction. This allows a plurality of workpieces that face each other to receive radiant heat from each other during transportation after heating to

provide supplemental amounts of heat to each other. Further, in addition to radiant heat, heat transfer also moves some amounts of heat from the workpiece to the ambient air. For example, if a plurality of workpieces are positioned to overlap in the direction perpendicular to the sheet surfaces of the workpieces while being in contact with each other, the contact portions are not in contact with air, which reduces amounts of heat moved through heat transfer from the workpieces to the air. Further, if a plurality of workpieces are positioned to overlap in the direction perpendicular to the sheet surfaces of the workpieces with some distance created between them, air warmed by heat transfer from the workpieces stays in the space between the heated workpieces. The presence of warmed air staying between the workpieces reduces the temperature differences between the air between the workpieces, on the one hand, and the workpieces themselves, on the other. This reduces movement of amounts of heat through heat transfer from the workpieces to the air, thus producing the temperature-retaining effect. This reduces temperature drop in the workpieces during transportation. The embodiments described below are based on these findings.

(22) A method of manufacturing a press-formed product according to an embodiment of the present invention includes: a heating step in which a sheet-shaped first workpiece and a sheet-shaped second workpiece are simultaneously heated; a transportation step in which the first and second workpieces heated at the heating step are transported to a press machine; and a pressing step in which the first and second workpieces transported to the press machine are processed by the press machine. At the heating step, the first and second workpieces are heated by a heating device while the first workpiece is placed on a first group of at least three struts extending upward from a tray body having an empty space extending therethrough in a top-bottom direction when viewed from above and the second workpiece is placed on a second group of at least three struts extending upward from the tray body and positioned above the first workpiece to overlap the first workpiece in a direction normal to a sheet surface of the first workpiece.

(23) The transportation step includes a first transportation sub-step and a second transportation sub-step. At the first transportation sub-step, the first and second workpieces, together with the tray body, are transported from the heating device to a lifting location while the first workpiece is placed on the first group of struts and the second workpiece is placed on the second group of struts and positioned above the first workpiece to overlap the first workpiece in the direction normal to the sheet surface of the first workpiece. At the second transportation sub-step, the second workpiece placed on the second group of struts, when at the lifting location, is lifted upward by a transportation device and transported to a press location for the second workpiece, and the first workpiece placed on the first group of struts, when at the lifting location, is lifted upward by the transportation device and transported to a press location for the first workpiece.

(24) In the above-described manufacturing method, the first and second workpieces are heated in the heating device while positioned by the groups of struts of the tray to overlap in the top-bottom direction. After heating, the first and second workpieces, together with the tray, are transported from the heating device to the lifting location. Thus, the first and second workpieces overlap in the top-bottom direction from when within the heating device until being lifted by the transportation device. That is, the first and second workpieces overlap in the direction perpendicular (i.e., normal) to the sheet surface of the first workpiece. For example, the upper surface of the first workpiece faces the lower surface of the second workpiece. This allows the plurality of workpieces to receive radiant heat from each other to provide supplemental amounts of heat to each other from completion of heating until lifting by the transportation device. Further, for example, if a plurality of workpieces are positioned to overlap with a distance, air warmed by heat transfer from the workpieces stays in the space between the workpieces. This produces the temperature-retaining effect. Furthermore, the first workpiece is placed on the first group of struts, and the second workpiece is positioned above it, placed on the second group of struts. The groups of struts extend upward from the tray body. Thus, the groups of struts do not present an obstacle when the second, and then first, workpieces placed on the groups of struts are lifted upward by the transportation

device. This enables performing the lifting operation in a simple and quick manner. As a result, in hot press-forming, temperature drop in the material during the transportation time after heating of the material until beginning of press-forming can be reduced in a simple manner.

(25) The tray body may be constructed such that the area of the empty space is larger than the total area occupied by the constituent elements of the tray body when viewed from above. This facilitates transmission of heat from below the tray body to the first and second workpieces during the heating step. Each of the struts of the first and second groups may be located between adjacent empty spaces of the tray body when viewed from above. This further facilitates transmission of heat from below the tray body through the empty spaces around the struts of the first and second groups to the first workpiece on the first group of struts and the second workpiece on the second group of struts.

(26) The sheet-shaped first and second workpieces are not limited to flat sheets. At least one of the sheet-shaped first and second workpieces may be, for example, a sheet with a projection protruding in the direction normal to the sheet surface, a sheet with an empty space extending through the sheet, or a curved sheet, for example. At least one of the first and second workpieces may be an intermediate formed product produced by forming or processing a flat sheet (i.e., blank), for example.

(27) It is preferable that, at the first transportation sub-step, the first workpiece with its lower surface supported by the first group of struts and the second workpiece with its lower surface supported by the second group of struts are transported while overlapping each other in the direction normal to the sheet surface of the first workpiece, separated by a distance. This allows the upper and lower surfaces of each of the first and second workpieces to be in contact with air during transportation. This reduces the differences between the conditions of the upper surface and those of the lower surface of each of the first and second workpieces during transportation compared with implementations where the workpieces are in contact with each other. This reduces the difference between the quality of the upper surface and that of the lower surface of each of the first and second workpieces.

(28) It is preferable that, at the heating step and the first transportation sub-step, the first workpiece placed on the first group of struts and the second workpiece placed on the second group of struts are positioned to overlap in the top-bottom direction so as to be separated by a distance not larger than 100 mm. This effectively reduces temperature drop by means of an appropriate distance between the first and second workpieces on the tray during transportation. If the distance between the heated workpieces is too large, an increased proportion of heat radiation, at the ends of the first and second heated workpieces, emitted from one workpiece in directions oblique to the normal to the sheet surface fails to reach the other (opposite) workpiece. As a result, the effect of reducing temperature drop may not be sufficiently produced. Further, if the distance between the workpieces is too large, air warmed by heat transfer from the two heated workpieces does not easily stay between the two workpieces, which means that the temperature-retaining effect may not be sufficiently produced.

(29) It is preferable that, at the heating step and the first transportation sub-step, a relationship between a maximum distance D (mm) between the first workpiece placed on the first group of struts and the second workpiece placed on the second group of struts as measured in the top-bottom direction, on one hand, and a minimum sheet thickness t (mm) of a thinnest portion of the first and second workpieces, on the other, is represented by the expression below. This effectively reduces temperature drop in the first and second workpieces during transportation.

$$D \leq 120t$$

(30) An area of a sheet surface of the second workpiece may be larger than an area of the sheet surface of the first workpiece. In such implementations, it is preferable that, at the heating step and the first transportation sub-step, the entire first workpiece placed on the first group of struts and the second workpiece placed on the second group of struts overlap in the top-bottom direction. This

allows the entire first workpiece to receive radiant heat from the second workpiece. This facilitates uniformly maintaining the temperature of the entire first workpiece. Further, the struts of the second groups may be positioned on portions of the second workpiece that do not overlap the first workpiece. The area of the sheet surface of the first workpiece is defined as the area of the first workpiece in top view as placed on the first group of struts. The area of the sheet surface of the second workpiece is defined as the area of the second workpiece in top view as placed on the second group of struts.

(31) A sheet thickness of the second workpiece may be larger than a sheet thickness of the first workpiece. That is, the first and second workpieces may have different sheet thicknesses. Further, in such implementations, temperature drop in the first workpiece with a smaller sheet thickness is reduced by radiant heat from the second workpiece with a relatively large heat capacity. This effectively reduces temperature drop in the first workpiece, which is thin and can easily experience temperature drop. This effect is particularly significant if the entire first workpiece and the second workpiece overlap in the top-bottom direction during transportation, as discussed above.

(32) It is preferable that, at the heating step and the first transportation sub-step, a minimum value of a distance ΔW between an edge of the second workpiece placed on the second group of struts as viewed from above and an edge of the first workpiece placed on the first group of struts as viewed from above is not smaller than 5 mm. This ensures that there are places where the second group of struts can be positioned while avoiding the first workpiece.

(33) It is preferable that, at the heating step and the first transportation sub-step, a maximum value of a distance ΔW between an edge of the second workpiece placed on the second group of struts as viewed from above and an edge of the first workpiece placed on the first group of struts as viewed from above is not larger than 30 mm. Thus, the area of the second workpiece that does not receive radiant heat from the first workpiece is not excessively large, which guarantees the effect of reducing temperature drop in the second workpiece.

(34) At the first transportation sub-step, the first workpiece placed on the first group of struts and the second workpiece placed on the second group of struts may be transported while being covered with a shield from a front as determined along a direction of transportation. The shield prevents outside air from hitting the front sides, along the direction of transportation, of the workpieces, thereby reducing temperature decrease beginning at the front sides, along the direction of transportation, of the workpieces and, at the same time, prevents air staying between the two workpieces, warmed by heat transfer from the workpieces, from moving to the outside. This retains the temperature-retaining effect. This further reduces temperature drop in the workpieces during transportation.

(35) The shield may include an inclined surface inclined so as to be located closer to the first and second workpieces going from a center toward an end. This allows air that has hit the shield to flow away from the first and second workpieces along the inclined surface. This increases the effect of reducing temperature drop in the workpieces during transportation.

(36) At least one of the first workpiece placed on the first group of struts and the second workpiece placed on the second group of struts may have a long-length direction and a short-length direction. At the first transportation sub-step, the shield can cover the first workpiece placed on the first group of struts and the second workpiece placed on the second group of struts along the long-length direction. Covering the workpieces with a shield along the long-length direction further increases the effect of reducing temperature drop in the workpieces during transportation.

(37) Each of the first and second workpieces may be a differential-thickness sheet with a large-thickness portion and a small-thickness portion. In such implementations, at the heating step and the first transportation sub-step, the large-thickness portion of the first workpiece placed on the first group of struts overlaps, as viewed in the top-bottom direction, the small-thickness portion of the second workpiece placed on the second group of struts. This allows the large- and small-thickness portions to provide supplemental amounts of heat to one another, thereby efficiently reducing

temperature drop as a whole.

(38) At the second transportation sub-step, the transportation device may simultaneously lift the second workpiece placed on the second group of struts and the first workpiece placed on the first group of struts, and transport the first and second workpieces from the lifting location to the press locations while the workpieces overlap in the direction normal to the sheet surface of the first workpiece. Simultaneous lifting of the first and second workpieces reduces the lifting time. Further, the first and second workpieces are transported from the lifting location to the press locations while overlapping each other. This further improves the effect of reducing temperature drop during transportation.

(39) By way of example, the second transportation sub-step may include: driving a pair of first arms rotatably mounted on a base frame included in the transportation device to support a lower surface, at both ends, of the first workpiece placed on the first group of struts using claws of the pair of first arms and lift the workpiece; driving, using a system separate from that for the pair of first arms, a pair of second arms rotatably mounted on the base frame of the transportation device to support a lower surface, at both ends, of the second workpiece placed on the second group of struts using claws of the pair of second arms and lift the workpiece; transporting the first workpiece with the lower surface supported, at both ends, by the claws of the pair of first arms of the transportation device and the second workpiece with the lower surface supported, at both ends, by the claws of the pair of second arms, while the workpieces overlap each other in the direction normal to the sheet surface of the first workpiece; driving the pair of first arms to lower the first workpiece supported by the pair of first arms to a press location in the press machine; and driving, using the system separate from that for the pair of first arms, the pair of second arms to lower the second workpiece supported by the pair of second arms to a press location in the press machine.

(40) Thus, the second and first workpieces may be lifted simultaneously, and the first and second workpieces may be transported from the lifting location to the press locations while overlapping in the direction normal to the sheet surface of the first workpiece. It will be understood that the transportation device is not limited to an arrangement that simultaneously lifts the second and first workpieces and transporting these workpieces while the workpieces overlap. For example, the transportation device may include a first transportation sub-device that lifts and transports the second workpiece and a second transportation sub-device that lifts and transports the first workpiece.

(41) A tray according to an embodiment of the present invention includes: a tray body shaped to expand in a plane perpendicular to a top-bottom direction and including an empty space extending therethrough in the top-bottom direction; and a first group of struts and a second group of struts extending upward from the tray body. The first group of struts include at least three struts constructed to be capable of supporting a lower surface of a sheet-shaped first workpiece; and the second group of struts include at least three struts constructed to be capable of supporting a lower surface of a sheet-shaped second workpiece located above the first workpiece supported by the first group of struts. The first group of struts are disposed such that imaginary straight lines connecting the struts form at least one triangle when viewed from above. The second group of struts are disposed at locations different from those of the first group of struts when viewed from above and disposed such that imaginary straight lines connecting the struts form at least one triangle when viewed from above. A top of every one of the struts of the second group is located higher than a top of that one of the struts of the first group whose top is located lowest.

(42) In the above-described arrangement, the first and second workpieces are supported, above the tray, by the first and second groups of struts while overlapping in the top-bottom direction. For example, the first and second workpieces are supported on the tray while the upper surface of the first workpiece and the lower surface of the second workpiece face each other. Thus, the first and second workpieces may be heated by the heating device while being placed on the tray so as to overlap in the top-bottom direction and, with this state kept, transported from the heating device to

the lifting location. This allows the first and second workpieces to receive radiant heat from each other to provide supplemental amounts of heat to each other from completion of heating until lifting by the transportation device. Further, the group of struts extend upward from the tray body. Thus, the group of struts do not present an obstacle during lifting upward, by the transportation device, of the second and first workpieces placed on the group of struts. It enables performing the lifting operation in a simple and quick manner. As a result, in hot press-forming, temperature drop in the material can be reduced in a simple manner during the transportation time after heating of the material until beginning of press-forming. That is, a tray used in heating/transportation of material for hot press-forming is provided that reduces temperature drop in the material during the transportation time in a simple manner.

(43) The group of struts may be fixed to the tray body. That is, the group of struts may be fixedly mounted on the tray body. This simplifies the construction of the tray. On the contrary, if the heights of the individual groups of struts are variable relative to the tray body, a heat-resistant driving mechanism is required. In such implementations, the construction is complicated and requires increased manufacturing costs.

(44) The second group of struts may include at least three struts with tops located higher than tops of at least three respective struts of the first group by a predetermined height ΔH . In such implementations, the second workpiece supported by the second group of struts can be positioned at a location higher than the first workpiece supported by the first group of struts by the predetermined height ΔH .

(45) It is preferable that the predetermined height ΔH is equal to a maximum sheet thickness of the first workpiece plus 0 to 100 mm. This keeps the distance between the first and second workpieces placed on the tray and transported at an appropriate amount, thereby effectively reducing temperature drop.

(46) The tray may further include a shield adapted to cover, along a direction perpendicular to the top-bottom direction, a height range between a height of that one of the struts of the first and second groups whose top is located lowest and a height of that strut whose top is located highest. The shield further reduces temperature drop in the first and second workpieces placed on the tray and transported.

(47) The shield may have an inclined surface inclined so as to be located closer to the first and second groups of struts going from a center toward an end. This allows outside air that has hit the shield during transportation of the first and second workpieces placed on the tray to flow away from the first and second workpieces along the inclined surface. This increases the effect of reducing temperature drop in the first and second workpieces during transportation.

(48) The tray body may have a long-length direction and a short-length direction as viewed from above. The shield may cover the tray body along the long-length direction as viewed from above. Covering the tray body with a shield along the long-length direction further increases the effect of reducing temperature drop in the workpieces placed on the tray and transported.

(49) A hot-press manufacturing line according to an embodiment of the present invention includes: the tray; a heating device adapted to heat the first workpiece and the second workpiece placed on the tray; a transportation path provided between the heating device and a lifting location for the first and second workpieces to transport the tray from the heating device to the lifting location; a transportation device adapted to support, and lift upward, the first and second workpieces placed on the tray when at the lifting location; at least one press machine including at least two pairs of die parts; and a moving device adapted to move the transportation device between the lifting location and press locations each located between the die parts of one of the at least two pairs of die parts.

(50) The transportation device may include: a base frame movable in a lateral direction perpendicular to the top-bottom direction; a pair of first arms rotatably mounted on the base frame; a pair of second arms rotatably mounted on the base frame; a first drive unit adapted to drive the first arms; and a second drive unit adapted to drive the second arms.

- (51) The pair of first arms include: a pair of first bases arranged in the lateral direction of the base frame and extending from the base frame in the top-bottom direction; and a pair of first claws extending, with a bend, from the respective first bases in the lateral direction.
- (52) The pair of second arms include: a pair of second bases arranged in the lateral direction of the base frame and extending from the base frame in the top-bottom direction; and a pair of second claws extending, with a bend, from the respective second bases in the lateral direction.
- (53) The first drive unit rotates the pair of first arms relative to the base frame to change a lateral distance between the first claws.
- (54) The second drive unit rotates the pair of second arms relative to the base frame to change a lateral distance between the second claws.
- (55) The first and second drive units are configured, independently from each other, to be capable of controlling rotation of the first arms and rotation of the second arms, respectively.
- (56) The pair of first claws are capable, when close to each other in the lateral direction, of supporting a lower surface of the first workpiece at both ends as determined along the lateral direction.
- (57) The pair of second claws are capable, when close to each other in the lateral direction, of supporting a lower surface of the second workpiece at both ends as determined along the lateral direction.
- (58) A location of the pair of first claws in the top-bottom direction and a location of the pair of second claws in the top-bottom direction are different from each other.
- (59) Now, embodiments of the present invention will be described in detail with reference to the drawings. The same or corresponding components in the drawings are labeled with the same reference numerals, and their description will not be repeated. The size ratios of the components shown in the drawings do not necessarily represent their actual size ratios.
- (60) [Exemplary Construction of Apparatus]
- (61) FIG. 1 is a schematic view of a hot-press manufacturing line **10** according to an embodiment. The hot-press manufacturing line **10** includes a heating device **14**, a transportation table **16**, a manipulator **44**, a transportation device **46**, a press machine **20**, and a controller **22**. The transportation table **16** is located near an outlet **14A** of the heating device **14**. The transportation table **16** is located between the heating device **14** and press machine **20**.
- (62) Inside the heating device **14**, a plurality of workpieces **W1** and **W2** are placed on a single tray **1** so as to overlap in the top-bottom direction. After the plurality of workpieces **W1** and **W2** placed on the tray **1** are heated by the heating device **14**, the workpieces, together with the tray **1**, are moved to exit the heating device **14**. The tray **1**, with the workpieces **W1** and **W2** placed thereon, exits the heating device **14** and is transported on the transportation table **16** to a location for lifting by the transportation device **46**.
- (63) (Heating Device)
- (64) The heating device **14** heats an object to be heated. The heating device **14** may be, for example, a resistance heating furnace, a gas heating furnace, a far-infrared heating furnace, or a near-infrared heating furnace. The heating device **14** is not limited to a heating furnace, and may be, for example, a high-frequency induction heating device, a low-frequency induction heating device, or an electrical heating device that causes electricity to directly flow through the object to be heated to heat the object. The heating device **14** may include a heating chamber. The heating device **14** may include, inside the heating chamber, a plurality of in-chamber rollers **13** that are driven by a driving mechanism, not shown, and rotate. Rotating the in-chamber rollers **13** transports the object being heated (i.e., workpieces **W1** and **W2**) on the in-chamber rollers **13**. The outlet **14A** and an inlet **14B** of the heating device **14** are located forward and rearward, respectively, as determined along the direction of transportation of the object being heated in the heating chamber.
- (65) (Transportation Table)

(66) The transportation table **16** includes a plurality of transportation rollers **26** that are driven by a driving mechanism, not shown, and rotate. As the transportation rollers **26** rotate in synchronization with the in-chamber rollers **13**, the object being transported is transported between the transportation table **16** and the interior of the heating chamber of the heating device **14**. The transportation rollers **26** are spaced apart from one another. The transportation table **16** is an example of a transportation path along which the tray **1** with the heated workpieces **W1** and **W2** placed thereon is transported from the heating device **14** to the lifting location. A lifting location for the workpieces by the transportation device **46** is provided above the transportation table **16**. It will be understood that the construction of the transportation path is not limited to the transportation table **16** shown in FIG. **1**. For example, the transportation path may be a belt conveyor, rails, or the like. Further, although the lifting location is on the transportation path in the implementation shown in FIG. **1**, the lifting location need not be on the transportation path. The lifting location may be on a stand, for example, provided separately from the transportation path and located at the destination of transportation.

(67) (Press Machine)

(68) The press machine **20** includes a lower die part **23** and an upper die part **21** for press-forming an object to be pressed. By way of example, the lower die part **23** is constituted by a punch; by way of example, the upper die part **21** is constituted by a die block. Channels for coolant may be provided in the upper and lower die parts **21** and **23**. Thus, heat taken from the object being pressed during press-forming may be released by means of coolant. Two workpieces may be positioned between the upper and lower die parts **21** and **23**. The upper and lower die parts **21** and **23** are movable relative to each other. With two workpieces positioned between the upper and lower die parts **21** and **23**, the press machine **20** moves the upper and lower die parts **21** and **23** closer to each other to press-form the two workpieces. The operation of the upper and lower die parts **21** and **23** may be controlled by the controller **22**, for example. In the present implementation, the lower and upper die parts **23** and **21** of the press machine **20** are shaped so as to simultaneously produce a plurality of press-formed products. In this implementation, two pairs of die parts are provided in one press machine. In the present implementation, a plurality of workpieces **W1** and **W2** are positioned between the lower and upper die parts **23** and **21** of the press machine **20** such that the plurality of workpieces **W1** and **W2** are press-formed simultaneously. A plurality of press machines **20** may be provided. For example, two press machines may be provided, each including one pair of die parts.

(69) (Manipulator)

(70) The manipulator **44** uses the transportation device **46** to transport the object being transported from the transportation table **16** to the press machine **20**. The transportation device **46** operates by lifting a workpiece, i.e., object being transported, holding it, and placing it on a surface. The manipulator **44** controls the location and attitude of the transportation device **46**. The transportation device **46** may be an end effector of the manipulator **44**. The manipulator **44** moves the transportation device **46** between the lifting location on the transportation table **16** and a press location between the die parts (i.e., upper and lower die parts **21** and **23**) of each of the two pairs of die parts in the press machine **20**. The manipulator **44** includes a base rotatable about at least one axis and an arm extending from the base and having at least one joint. The transportation device **46** is rotatably mounted on the distal end of the arm. A moving device for moving the transportation device **46** is not limited to a manipulator. For example, a moving device may include rails connecting the transportation table **16** and press machine **20** and a suspension device for suspending the transportation device **46** above the rails and moving it up and down.

(71) (Controller)

(72) The Controller **22** controls the heating device **14**, transportation table **16**, press machine **20**, and manipulator **44** (and thus transportation device **46**). The controller **22** is constituted by one or more computers each including a processor and memory, for example. The control of these

components is implemented by the processor executing a program stored on the memory. The controller **22** transmits control signals or control data to the heating device **14**, transportation table **16**, press machine **20**, and manipulator **44** (and thus transportation device **46**). Further, the controller **22** may receive signals or data indicating the states of these components. By way of example, the controller **22** may include control units (constituted by, for example, circuits or processors) distributed among the heating device **14**, transportation table **16**, manipulator **44**, and press machine **20** for controlling these devices. In such implementations, the controller **22** may include an overall-control computer that supplies the control units of the various devices with control information to control the operation of the entire hot-press manufacturing line **10**. The operation of the arms **71** of the transportation device **46**, described further below, may be controlled by part of the controller **22**, such as the control unit of the manipulator **44**.

(73) (Transportation Device)

(74) The transportation device **46** includes a base frame **48**, as well as at least one pair of arms **71** rotatably mounted on the base frames **48**. A joint **56** is provided on the upper surface of the base frame **48** to be connected to the manipulator **44**. The joint **56** is connected to the manipulator such that the base frame **48** is rotatable, about an axis in the top-bottom direction, relative to the manipulator **44**.

(75) The pair of arms **71** are separated from each other in a lateral direction of the base frame **48** (i.e., direction perpendicular to the top-bottom direction). Each of the arms **71** includes a base extending from the base frame **48** in the top-bottom direction, and a claw extending, with a bend, from the base in the lateral direction. Controlling rotation of the arms **71** relative to the base frame **48** controls the arms to be in a closed state, where the claws of the arms **71** are located closer to each other, and an open state, where they are located far away from each other.

(76) (Tray)

(77) FIG. **2** is a top view of the tray **1**. FIG. **3** is a side view of the tray **1** shown in FIG. **2** as viewed in the direction of arrow F. The tray **1** is a tray used in heating/transporting of the material (i.e., workpiece) for hot pressing. The tray **1** is constructed to resist heating temperatures for workpieces. In the implementation shown in FIG. **2**, the tray body **2** includes empty spaces **2G** shaped to expand in a plane perpendicular to the top-bottom direction and extending in the top-bottom direction through the tray body. When viewed from above, the total area of the empty spaces **2G** is larger than the total area occupied by the constituent members of the tray body **2**. The tray **1** includes a plurality of struts **3** extending upward from the tray body **2**. The plurality of struts **3** include a first group of struts **3a** that allow the first workpiece W**1** to be placed thereon, and a second group of struts **3b** that allow the second workpiece W**2** to be placed thereon above the first workpiece W**1**. Every one of the struts of the first and second groups **3a** and **3b** is located between adjacent empty spaces **2G** of the tray body **2** as viewed from above.

(78) (Tray Body)

(79) In the implementation shown in FIG. **2**, the tray body **2** includes a frame **2c** and rod members **2f** located inside the frame **2c** and bridging the frame. The frame **2c** includes a pair of vertical frame members **2b** and a pair of horizontal frame members **2a**. The pair of vertical frame members **2b** are located parallel to each other, and laterally separated. The pair of horizontal frame members **2a** are located between the vertical frame members **2b** and parallel to each other, and vertically separated. The pair of vertical frame members **2b** and the pair of horizontal frame members **2a** form a rectangular frame **2c** when viewed from above. The rod members **2f** include vertical rod members **2d** and horizontal rod members **2e**. The vertical rod members **2d** bridge the pair of horizontal frame members **2a**. The horizontal rod members **2e** bridge the pair of vertical frame members **2b**. The rod members **2f** are disposed grid-wise inside the frame **2c**.

(80) The rod members **2f** (i.e., at least one of the set of vertical rod members **2d** and the set of horizontal rod members **2e**) may be constructed to be adjustable in position relative to the frame **2c**. For example, the frame **2c** may be provided with a plurality of positioning holes or engagement

lugs. In such implementations, the rod members **2f** are fixed to the holes or engagement lugs on the frame **2c** using fastening members, for example, as necessary. The positions of the rod members **2f** relative to the frame **2c** can be adjusted by changing the holes or engagement lugs, i.e. locations, to which the rod members **2f** are fixed.

(81) The construction of the tray body **2** is not limited to the implementation shown in FIG. 2. For example, the tray body may have the shape of a ladder including a pair of vertical rod members disposed generally parallel and separated from each other, and a plurality of horizontal rod members located between the vertical rod members and bridging the pair of vertical rod members in the direction perpendicular thereto. Alternatively, the tray body may be constituted by a sheet-shaped member including a plurality of holes extending through the tray body in the top-bottom direction, which serve as empty spaces.

(82) The constituent members of the tray body **2** (in the implementation of FIG. 2, frame **2c** and rod members **2f**) may be pipes or solid members. Alternatively, the constituent members of the tray body **2** may be angle members with an L-shaped cross section, or channel members with a U-shaped cross section. The constituent members of the tray body **2** are not limited to any particular material, and may be formed from a heat-resistant material, such as heat-resistant steel or ceramics, for example. It is desirable that the maximum use temperature of the constituent members be, for example, not lower than 900° C., a temperature range commonly found in heating devices, and not higher than 1050° C., which is an upper-limit setting temperature for heating devices. Examples of heat-resistant steels (heat-resistant alloy steels) that can be used for the constituent members include SCH22 (0.4C-25Cr-20Ni), SCH24 (0.4C-25Cr-35Ni—Mo, Si). Constituent members of the tray body **2** formed from a heat-resistant alloy steel can be easily processed and produced.

(83) (Struts)

(84) The first group of struts **3a** include at least three struts, where the imaginary straight lines connecting the struts form a triangle when viewed from above. The second group of struts **3b** include at least three struts, where the imaginary straight lines connecting the struts form a triangle when viewed from above. Every one of the struts of the second group **3b** is at a location different from that of any one of the struts of the first group **3a** when viewed from above. The top of every one of the struts of the second group **3b** is located higher than the top of that strut of the first group **3a** whose top is located lowest. Thus, the first group of struts **3a** are capable of supporting the first workpiece **W1**. The second group of struts **3b** are capable of supporting the second workpiece **W2** such that this workpiece is located above the first workpiece **W1** supported by the first group of struts **3a**.

(85) The second group of struts **3b** are positioned in the sub-regions occupied by the second workpiece **W2** when viewed from above but that do not overlap the region occupied by the first workpiece **W1**. Further, the second group of struts **3b** are constructed not to overlap the region occupied by the first workpiece **W1** when viewed from above. That is, the second group of struts **3b** are constructed such that the first workpiece **W1**, when lifted up by the transportation device **46**, does not get snagged on the second group of struts **3b**.

(86) The struts of the first group **3a** and those of the second group **3b** are not limited to any particular numbers. The number of the struts of the first group **3a** and the number of the struts of the second group **3b** may be the same or different. For example, the number of the struts of the second group **3b** may be larger than the number of the struts of the first group **3a** if the support for the second workpiece **W2** is to be stronger than that for the first workpiece **W1** to take account of the characteristics and/or support locations of the first and second workpieces **W1** and **W2**, for example.

(87) The struts of the first and second groups **3a** and **3b** (hereinafter simply referred to as struts **3** when no distinction is made) may each have the shape of a column, such as a circular column or a rectangular column, or a cone, such as a circular cone or a pyramid, for example. Each strut **3** may be a solid member or a pipe. The struts **3** are not limited to any particular material, and may be

formed from a heat-resistant material, such as heat-resistant steel or ceramics, for example. It is desirable that the maximum use temperature of the struts **3** be, for example, not lower than 900° C., a temperature range commonly found in heating devices, and not higher than 1050° C., which is an upper-limit setting temperature for heating devices. Examples of heat-resistant steels (heat-resistant alloy steels) that can be used for the constituent members of the struts **3** include SCH22 (0.4C-25Cr-20Ni) and SCH24 (0.4C-25Cr-35Ni—Mo, Si). The struts **3** are fixed to the tray body **2**. For example, the struts **3** may be inserted into holes in the tray body **2** and thus attached to the tray body **2**. In such implementations, for example, the struts **3** may be fixed to the tray body **2** by a male thread provided on the outer periphery of an end of a strut **3** engaging a female thread on the inner periphery of a hole in the tray body **2**. Alternatively, the struts **3** may be press-fit into holes in the tray body **2**. The means for fixing the struts **3** are not limited to holes in the tray body **2**. The struts **3** may be fixed to the tray body **2** by other fixing means.

(88) (Exemplary Arrangement of First and Second Workpieces)

(89) In the implementation shown in FIG. 2, the first workpiece W1 includes notches on edges (i.e., ends) when viewed from above. The struts of the second group **3b** are located in regions corresponding to the notches of the first workpiece W1. Thus, as the struts of the second group **3b** are located in the regions corresponding to the notches or holes of the first workpiece W1, the struts of the second group **3b** can be positioned in regions that do not overlap the first workpiece W1 when viewed from above. For example, at least one of the struts of the second group **3b** may be positioned in a region corresponding to a locating hole of the first workpiece W1. A locating hole is used to position the first workpiece W1 during press-forming.

(90) FIG. 4 is a top view of a tray with another exemplary arrangement of the first and second groups **3a** and **3b** of struts. FIG. 5 is a side view of the tray **1** shown in FIG. 4 as viewed in the direction of arrow F. In the implementation shown in FIGS. 4 and 5, the first and second workpieces W1 and W2, which have the same shape, are positioned to be displaced from each other as viewed from above. That is, when viewed from above, the first and second workpieces W1 and W2 are positioned such that some portions of the second workpiece W2 do not overlap the first workpiece W1. The struts of the second group **3b** are disposed at locations in a region occupied by the second workpiece W2 and that do not overlap the region occupied by the first workpiece W1. In such implementations, no notches or holes need be provided in the first workpiece W1. In the implementation shown in FIG. 4, the shape of the first workpiece W1 as viewed from above results from a 180-degree rotation of the second workpiece W2 about an axis in the direction normal to the sheet surfaces.

(91) In the implementations shown in FIGS. 2 to 5, the difference between the height (i.e., length as measured in the top-bottom direction) of the second group of struts **3b** and that of the first group of struts **3a** is denoted by ΔH . Thus, at least three pairs of struts may be provided, each pair having a strut of the first group **3a** and another strut of the second group **3b** whose top is higher than the top of the former by ΔH . This allows the first and second workpieces W1 and W2 to be positioned so as to overlap in the top-bottom direction, with a distance of ΔH created between the lower surface of the first workpiece and the lower surface of the second workpiece. For example, if both the first and second workpieces W1 and W2 are flat sheets as in the implementations of FIGS. 2 to 5, all the struts of the first group **3a** have the same height, and all the struts of the second group **3b** have the same height. Thus, the top of every one of the struts of the second group **3b** is located higher than the top of the corresponding one of the struts of the first group **3a** by ΔH . If ΔH is equal to the sheet thickness of the first workpiece W1, the upper surface of the first workpiece W1 and the lower surface of the second workpiece W2 are in contact.

(92) The first and second workpieces W1 and W2 need not be flat sheets. Each of FIGS. 6 and 7 is a side view of a tray with groups of struts accommodating first and second workpieces that are not flat sheets. FIG. 6 shows an implementation for first and second workpieces W1 and W2 that are intermediate formed products with the same shape. In this implementation, each of the first and

second workpieces W1 and W2 has been processed to have the same hat-shaped cross section. In this implementation, too, at least three pairs of struts are provided, each pair having a strut of the first group 3a and another strut of the second group 3b whose top is higher than the top of the former by ΔH . ΔH is the distance between the lower surface of the first workpiece W1 and the lower surface of the second workpiece W2 as measured in the top-bottom direction.

(93) As shown in FIG. 7, the first and second workpieces W1 and W2 may have different cross-sectional shapes. In this implementation, too, at least three pairs of struts are provided, each pair having a strut of the first group 3a and another strut of the second group 3b whose top is higher than the top of the former by ΔH .

(94) The distance between the first and second workpieces W1 and W2 placed on the tray 1 is determined by the distance ΔH between the heights of the struts of the first group 3a and the heights of the corresponding struts of the second group 3b. That is, the distance D in the top-bottom direction between the first and second workpieces W1 and W2 on the tray 1 during heating and transportation is determined by ΔH . The maximum value of the distance D between the first and second workpieces W1 and W2 on the tray during heating and transportation is preferably not larger than 100 mm, for example. The smaller the distance D, the better to allow the first and second workpieces W1 and W2 on the tray 1 to receive radiant heat from each other to reduce temperature drop. The maximum value of the distance D is more preferably not larger than 50 mm, yet more preferably not larger than 30 mm, still more preferably not larger than 10 mm. No particular lower limit for the distance D is specified; the distance required for the operation of placing the workpieces on the tray 1 and lifting them therefrom represents the lower limit for the distance D. A preferred range for ΔH is a preferred range for the distance D plus the maximum sheet thickness of the first workpiece W1. For example, if the first workpiece W1 has a sheet thickness of about 3 mm, a preferred range for ΔH may be 3 to 103 mm.

(95) FIG. 8 is a top view of a tray 1 accommodating first and second workpieces W1 and W2 with different surface areas as viewed from above. FIG. 9 is a side view of the tray shown in FIG. 8 as viewed in the direction of arrow F. In the implementation shown in FIGS. 8 and 9, the area of a sheet surface of the second workpiece W2 is larger than the area of a sheet surface of the first workpiece W1. Thus, the first and second workpieces W1 and W2 may have different sizes. The second workpiece W2 overlaps the entire first workpiece W1 in the top-bottom direction. Thus, as viewed from above, an end (i.e., edge) of the second workpiece W2 is located outward of the corresponding end (i.e., edge) of the first workpiece W1. The second group of struts 3b are located in the regions between the edges of the second workpiece W2 and the edges of the first workpiece W1 as viewed from above. That is, the second group of struts 3b are located in the regions where the second workpiece W2 does not overlap the first workpiece W1 as viewed from above. Thus, the first and second workpieces W1 and W2 may be supported by the first and second groups of struts 3a and 3b so as to overlap each other in the top-bottom direction even without notches or holes, for example, in the first workpiece W1. In such implementations, the second group of struts 3b are located in the regions outside the regions where the first group of struts 3a are located as viewed from above.

(96) The distance ΔW between an edge of the second workpiece W2 placed on the second group of struts 3b as viewed from above and the corresponding edge of the first workpiece W1 placed on the first group of struts 3a as viewed from above is not limited to any particular value, but is preferably not larger than 30 mm. If ΔW is too large, it means a large area of edge portions of the second workpiece W2 that do not overlap the first workpiece W1. This decreases the effect of reducing temperature drop in the edge portions of the second workpiece W2. In view of this, ΔW is more preferably not larger than 20 mm, and yet more preferably not larger than 15 mm. If ΔW is too small, it is difficult to ensure that there is an area where the second group of struts 3b can be disposed. In view of this, the minimum value of ΔW is preferably not smaller than 5 mm, more preferably not smaller than 8 mm, and yet more preferably not smaller than 10 mm.

(97) Here, ΔW is the distance between an edge of the second workpiece W2 and the corresponding edge of the first workpiece W1 as measured in the direction perpendicular to the line of the edge of the second workpiece W2 (if the line of the edge is a curved line, the direction normal thereto) as viewed from above.

(98) In FIGS. 8 and 9, both the first and second workpieces W1 and W2 are flat sheets; alternatively, at least one of the first and second workpieces W1 and W2 may be an intermediate formed product in lieu of a flat sheet. In such implementations, too, the area of the second workpiece W2 placed on the second group of struts 3b may be larger than the area of the first workpiece W1 placed on the first group of struts 3a as viewed from above.

(99) In the implementation shown in FIG. 9, the sheet thickness of the second workpiece W2 is larger than the sheet thickness of the first workpiece W1. The smaller the sheet thickness, the higher the rate of temperature drop. Thus, positioning the entire first workpiece W1, which has a smaller sheet thickness, so as to overlap the second workpiece W2, which has a larger sheet thickness, increases the effect of reducing temperature drop in the first workpiece W1. As a result, the effect of reducing temperature drop in the first and second workpieces W1 and W2 as a whole is increased. Alternatively, the first and second workpieces W1 and W2 may have the same sheet thickness.

(100) (Exemplary Arrangements of Differential-Thickness Sheets)

(101) At least one of the first and second workpieces W1 and W2 may be a differential-thickness sheet including a large-thickness portion and a small-thickness portion. FIG. 10 is a side view of a tray accommodating first and second workpieces W1 and W2 that are differential-thickness sheets. In the implementation shown in FIG. 10, the large-thickness portion of the first workpiece W1 placed on the first group of struts 3a and the small-thickness portion of the second workpiece W2 placed on the second group of struts 3b overlap each other in the top-bottom direction. This allows the two workpieces to be transported while the small-thickness portion of one workpiece, which has a smaller sheet thickness within this workpiece and thus can experience temperature drop more easily, is faced by the large-thickness portion of the other workpiece, which has a relatively high heat capacity within this workpiece. This reduces temperature drop in the small-thickness portions of both workpieces more effectively. This efficiently reduces temperature drop in the entirety of both workpieces and, at the same time, efficiently reduces the temperature difference between the small-thickness portion and large-thickness portion of each workpiece.

(102) For example, if the first and second workpieces W1 and W2 have the same shape and have small- and large-thickness portions in the same distribution, the first workpiece W1 may be placed on the first group of struts 3a, and then the second workpiece W2 may be placed on the second group of struts 3b in such a manner that the orientation of the second workpiece W2 has been rotated about an axis in the top-bottom direction (e.g., by 180 degrees) relative to the first workpiece W1 so as to be in a different orientation therefrom. To do this, the orientation, i.e., amount of rotation, of the second workpiece W2 may be adjusted such that at least part of the large-thickness portion of the first workpiece W1 overlaps at least part of the small-thickness portion of the second workpiece W2 as viewed from above.

(103) A differential-thickness sheet may be, for example, a tailored blank produced by joining steel sheets with different sheet thicknesses with adjacent ends abutting each other. Alternatively, a differential-thickness sheet may be a patchwork-tailored blank produced by joining overlapping steel sheets with different sizes. Alternatively, a differential-thickness sheet may be a tailor-rolled blank made up of one steel sheet with some portions with sheet thicknesses changed through processing, such as rolling.

(104) (Shields)

(105) FIG. 11 is a top view of an exemplary arrangement of the tray 1 including shields 4 (4A to 4D). FIG. 12 is a side view of the tray 1 shown in FIG. 11 as viewed in the direction of arrow F. The tray 1 shown in FIGS. 11 and 12 is the same as the tray 1 shown in FIGS. 1 and 2 except that

the shields **4** (**4A** to **4D**) have been added. As shown in FIG. **11**, the shields **4A** to **4D** are provided so as to surround the entire periphery of the tray body **2** as viewed from above. The shields **4** include a pair of shields **4A** and **4C** covering the tray body **2** along the short-length direction (i.e. short sides) and a pair of shields **4B** and **4D** covering the tray body along the long-length direction (i.e., long sides) as viewed from above.

(106) As shown in FIG. **12**, every one of the shields **4A** to **4D** covers the height region defined by the heights of the lowest-top strut and the highest-top strut of the first and second groups **3a** and **3b** from a side (i.e., in a direction perpendicular to the top-bottom direction). That is, the shields **4A** to **4D** extend from a location lower than the top of the lowest-top strut to a location higher than the top of the highest-top strut. Thus, the shields **4A** to **4D** prevent outside air from flowing into the space between the first workpiece **W1** placed on the first group of struts **3a** and the second workpiece **W2** placed on the second group of struts **3b**.

(107) The shields **4A** to **4D** are mounted on the tray body **2**. In the implementation shown in FIG. **12**, the shields **4B** and **4D** are connected to the sides of the tray body **2** and extend upward. The shields **4** are connected to the tray body **2** by means of fastening members such as welds or bolts, for example. The shields **4** may be detachable from the tray body **2**.

(108) In the implementation shown in FIGS. **11** and **12**, each of the shields **4A** to **4D** has an inclined surface inclined so as to be located closer to the struts **3** and thus the first and second workpieces **W1** and **W2** going from the center to the ends. In other words, each of the shields **4A** to **4D** is curved in shape such that its central portion protrudes outward and its ends are located inward of the central portion. Thus, outside air that has hit the shields **4** is guided away from the first and second workpieces **W1** and **W2** placed on the tray. As shown in FIG. **12**, the shields **4B** and **4D** are constructed such that their central portions are located outward of their ends as viewed both from above and from the sides.

(109) It will be understood that the shields **4** are not limited to the above-described shape. For example, the shields **4** may be constituted by flat sheets. The shields may also be shaped such that their central portions are located outward of their ends as viewed either in side view or in plan view.

(110) Also, the positioning of the shields **4** are not limited to the locations shown in FIGS. **11** and **12**. In the implementation shown in FIG. **11**, shields **4B** and **4D** covering the workpieces along the long length and shields **4A** and **4C** covering the workpieces along the short length are provided. In other arrangements, shields **4** may cover the workpieces along at least one of the long length and the short length. Covering the workpieces along the long length with shields **4** prevents air from entering in a wide range. In other arrangements, a shield **4** covers the workpieces at least from the front as determined along the direction of transportation. This prevents outside air from entering during transportation more effectively.

(111) [Exemplary Process for Manufacturing Press-Formed Product]

(112) Returning to FIG. **1**, an exemplary process for manufacturing a press-formed product using the above-described hot-press manufacturing line **10** will be described. The process for manufacturing a press-formed product according to an embodiment includes a heating step in which the first and second workpieces **W1** and **W2** are heated, a transportation step in which the first and second workpieces **W1** and **W2** are transported, and a pressing step in which the first and second workpieces **W1** and **W2** are pressed.

(113) (Heating Step)

(114) At the heating step, the first and second workpieces **W1** and **W2** are heated in the heating device **14** while being placed on the tray **1**. For example, as shown in FIGS. **2** to **12**, the first workpiece **W1** is placed on the first group of struts **3a** and the second workpiece **W2** is placed on the second group of struts **3b**. The first and second workpieces **W1** and **W2** placed on the tray **1** overlap each other in the top-bottom direction, i.e., direction normal to the sheet surfaces of the first workpiece **W1**.

(115) The tray body **2** includes empty spaces **2G** extending therethrough in the top-bottom direction. Thus, even if a heat source is located below the tray **1**, heat transfer from the heat source to the first and second workpieces **W1** and **W2** is less likely to be prevented by the tray body **2**.

(116) The tray **1** with the first and second workpieces **W1** and **W2** placed thereon enters the heating chamber of the heating device **14** through the inlet **14B**. The tray **1** is transported by the in-chamber rollers **13** to the outlet **14A**. The heating device **14** heats the first and second workpieces **W1** and **W2** to a predetermined heating temperature while transporting the workpieces. This heating temperature is, for example, the Ac3 transformation point, at which the ferrite in the steel constituting the first and second workpieces **W1** and **W2** completes transformation to austenite, or higher.

(117) It will be understood that a transportation device and a manipulator may be provided for placing the first workpiece **W1** and then second workpiece **W2** on the tray **1** before they enter the heating device **14** such that the workpieces overlap each other. The transportation device places the first workpiece **W1** on the first group of struts **3a** of the tray **1** and, thereafter, places the second workpiece **W2** on the second group of struts **3b**. For example, this transportation device may have the same construction as the transportation device **46** for lifting the first and second workpieces **W1** and **W2** and placing them on the press locations shown in FIG. **1**. This transportation device is located upstream of the heating device **14**.

(118) (First Transportation Sub-Step)

(119) At a first transportation sub-step, the first and second workpieces **W1** and **W2** exit the heating device **14** while being placed on the tray **1**, and are transported to a location where they are handed to the transportation device **46**, i.e., lifting location. Thus, the workpieces, together with the tray **1**, are transported from the heating device **14** to the lifting location while the first workpiece **W1** is placed on the first group of struts **3a** and the second workpiece **W2** is placed on the second group of struts **3b** and overlaps the first workpiece **W1**, located above it. Outside the heating device **14**, the tray **1** is transported by the transportation rollers **26** to near the press machine **20**. For example, as illustrated by FIGS. **2** to **12**, the first and second workpieces **W1** and **W2** are placed on the tray **1** and transported from the heating device **14** to the lifting location. The positional relationship between the first and second workpieces **W1** and **W2** at the first transportation sub-step is the same as their positional relationship at the heating step. That is, the tray **1** makes it possible to transport the first and second workpieces **W1** and **W2** to the lifting location at the first transportation sub-step while retaining the positional relationship between the first and second workpieces **W1** and **W2** during heating. This reduces temperature drop in the first and second workpieces **W1** and **W2** during transportation. Further, temperature drop in the workpieces during transportation is reduced even without providing a heat source, such as a heater, on the transportation path to generate heat energy. Thus, temperature drop is reduced in a simple manner.

(120) It will be understood that, after the first and second workpieces **W1** and **W2** have been lifted by the transportation device **46** and transported to the press locations, the tray **1** may be used for the heating and transportation sub-steps for other workpieces.

(121) (Second Transportation Sub-Step)

(122) At a second transportation sub-step, near the press machine **20**, the second workpiece **W2** placed on the second group of struts **3b** is lifted upward by the transportation device **46** and, thereafter, the first workpiece **W1** placed on the first group of struts **3a** is lifted by the transportation device **46**. As discussed above, the transportation device **46** moves the first workpiece **W1** upward while holding the workpiece with the arms **71**. The first workpiece **W1** moves upward while remaining in the same attitude as when being placed on the first group of struts **3a**. At this time, the first workpiece **W1** being lifted upward by the transportation device **46** does not get snagged on the second group of struts **3b**, since the second group of struts do not overlap the first workpiece **W1** placed on the first group of struts **3a** as viewed from above. This simplifies the control or device configuration for lifting the first workpiece **W1**.

(123) It will be understood that the transportation device for lifting the second workpiece W2 and the transportation device for lifting the first workpiece W1 need not necessarily be the same device. The first and second workpieces W1 and W2 lifted by the transportation device 46 are set on their respective press locations. The press locations are between the upper die part 21 and lower die part 23 of the press machine 20, for example.

(124) The first and second workpieces W1 and W2 may be set on the same press location, or on different press locations. Further, the first and second workpieces W1 and W2 may be set on different locations in the die of a single press machine 20. Alternatively, two press machines 20 may be provided. In such implementations, the first workpiece W1 may be set in the die of one of the two press machines 20, whereas the second workpiece W2 may be set in the die of the other press machine.

(125) (Pressing Step)

(126) The press machine 20 press-forms the first and second workpieces W1 and W2 using the upper and lower die parts 21 and 23. The first and second workpieces W1 and W2 being press-formed here have been heated by the heating device 14 to the Ac3 transformation point or higher and then transported to the press locations by the transportation table 16 and transportation device 46. The press machine 20 begins press-forming the first and second workpieces W1 and W2 using the upper and lower die parts 21 and 23 to perform forming and quenching at the same time. Specifically, with the first and second workpieces W1 and W2 being positioned between the upper and lower die parts 21 and 23, the press machine 20 moves the upper and lower die parts 21 and 23 closer to each other until the die reaches the bottom-dead point to form the workpieces. After the die has reached the bottom-dead point, the upper and lower die parts 21 and 23, while being die-tightened, are in contact with the first and second workpieces W1 and W2 to rapidly remove heat from the first and second workpieces W1 and W2. This causes martensite transformation or bainite transformation of the first and second workpieces W1 and W2. As a result, the first and second workpieces W1 and W2 are made into formed products that have been shaped to conform to the upper and lower die parts 21 and 23 and quenched.

(127) The present invention is not limited to the above-described embodiments. For example, although the above-described embodiments describe implementations with two workpieces (i.e., first and second workpieces W1 and W2) placed on the tray 1 while overlapping each other in the top-bottom direction, three or more workpieces may be placed on the tray 1 while overlapping one another in the top-bottom direction.

(128) [Material of Workpieces]

(129) The material of a workpiece may be any formable metal. Although not limiting, examples of materials for a workpiece include Fe-based materials such as carbon steel or stainless steel, and Al-based and Ti-based materials. Further, the workpiece may include a plating layer. For example, the workpiece may be a plated steel sheet. The plating layer may be a plating layer of, for example, an aluminum alloy, an aluminum-based alloy, a zinc alloy, or a zinc-based alloy.

(130) A workpiece constituted by a plated steel sheet is preferable in that oxide scale is not easily produced during transportation. If two workpieces W1 and W2 overlapping in the top-bottom direction are transported on the tray 1 and, during this, oxide scale is formed on the lower surface of the upper second workpiece W2, this oxide scale may peel off and drop onto the upper surface of the lower first workpiece W1. Further, oxide scale formed on the lower surface of the lower first workpiece W1 may peel off, too. In such cases, the difference between the amounts of oxide scale adhering to the upper and lower surfaces of the lower first workpiece W1 may be larger than the corresponding difference for the upper second workpiece W2. A difference in the amount of adhering scale may lead to differences in frictional characteristics of the die and workpiece surface during press-forming. As a result, separate adjustments to the die and separate settings of forming conditions may be needed for each workpiece. If the workpiece is a plated steel sheet, oxide scale formed during transportation can be reduced, thereby reducing variations in characteristics of

workpieces.

(131) [Effects of Transportation of Plurality of Workpieces Separated with Distance]

(132) In the above-described implementations, first and second workpieces **W1** and **W2** on the tray **1**, separated in the top-bottom direction and overlapping each other, are transported. This reduces variations in characteristics of a workpiece as represented by the differences between the upper and lower surfaces, and variations in characteristics of the upper and lower surfaces as represented by the differences between the upper and lower workpieces.

(133) If the first and second workpieces **W1** and **W2** overlapping with no clearance therebetween are transported, the overlying surfaces of the workpieces, on the one hand, and the opposite surfaces of the workpieces, on the other, are in contact with air for different periods of time. In such cases, if the first and second workpieces **W1** and **W2** are non-plated sheets, the upper and lower surfaces of each workpiece may develop different amounts of oxide scale. If the upper and lower surfaces have different amounts of oxide scale, the upper and lower surfaces may have different frictional characteristics. Further, the overlying surface and the opposite surface of the first workpiece **W1** are in reversed up-down orientations compared with those of the second workpiece **W2**; thus, the first and second workpieces **W1** and **W2** themselves may have different frictional characteristics. As a result, adjustments to the die and settings of forming conditions for press-forming may require increased time. Further, if first and second workpieces **W1** and **W2** that are sheeted steel sheets are heated at the heating step while overlapping each other without a clearance between the workpieces, the overlying surface and the opposite surface of each workpiece may experience different rates of temperature increase, potentially causing differences in temperature history. If this is the case, the upper and lower surfaces of each workpiece may experience differences in the alloying of the plating and base material, potentially causing differences in the quality of the plating. In view of this, the first and second workpieces **W1** and **W2** are separated by a distance while being heated and transported to reduce variations in characteristics, such as frictional characteristics and plating quality, of the upper and lower surfaces of the workpieces.

(134) [Variations of Hot-Press Manufacturing Line]

(135) FIG. **13** shows a variation of the hot-press manufacturing line **10**. The hot-press manufacturing line **10** shown in FIG. **13** includes a heating device **14**, a transportation table **16**, a manipulator **44**, a transportation device **46**, a press machine **20**, and a controller **22**. The hot-press manufacturing line **10** of FIG. **13** may have the same configuration as in FIG. **1** except for the transportation device **46**.

(136) (Transportation Device)

(137) FIG. **14A** is a side view of the transportation device **46** of FIG. **13** as viewed from a side (y-direction) illustrating an exemplary construction. FIG. **14B** shows the transportation device **46** of FIG. **14A** with its first arms **71** opened outwardly.

(138) (Base Frame)

(139) In the implementation shown in FIGS. **14A** and **14B**, the transportation device **46** includes a base frame **48**, as well as a pair of first arms **71** and a pair of second arms **72** rotatably mounted on the base frames **48**. The shape of the base frame **48** is rectangular when viewed from above. In the present implementation, the top-bottom direction represents the z-direction. A lateral direction is a direction in a plane perpendicular to the top-bottom direction. Lateral directions include the long-length direction of the base frame **48**, i.e., y-direction, and the short-length direction, i.e., x-direction.

(140) A joint **56** is provided on the upper surface of the base frame **48** to be connected to the manipulator **44**. The joint **56** is connected to the manipulator such that the base frame **48** is rotatable, about an axis in the top-bottom direction, relative to the manipulator **44**.

(141) (First and Second Arms)

(142) The pair of first arms **71** are separated from each other in a lateral direction (x-direction). Each of the first arms **71** includes a first base **71a** extending from the base frame **48** in the top-

bottom direction, and a first claw **71b** extending, with a bend, from the first base **71a** in the lateral direction. Each first arm **71** is mounted on the base frame **48** so as to be rotatable about an axis of rotation **60** in the y-direction. One end of the first base **71a** is rotatably connected to the base frame **48**, while the first claw **71b** extends from the other end.

(143) The pair of second arms **72** are separated from each other in a lateral direction (x-direction). Each of the second arms **72** includes a second base **72a** extending from the base frame **48** in the top-bottom direction, and a second claw **72b** extending, with a bend, from the second base **72a** in the lateral direction. Each second arm **72** is mounted on the base frame **48** so as to be rotatable about an axis of rotation **60** in the y-direction. One end of the second base **72a** is rotatably connected to the base frame **48**, while the second claw **72b** extends from the other end.

(144) The location of the pair of first claws **71b** in the top-bottom direction and the location of the pair of second claws **72b** in the top-bottom direction are different from each other. In the implementation shown in FIGS. **14A** and **14B**, the first bases **71a** are longer than the second bases **72a** as measured in the top-bottom direction. The first claws **71b** are located more distant from the base frame **48** than the second claws **72b** are.

(145) In the implementation shown in FIGS. **14A** and **14B**, the axis of rotation **60** of each first arm **71** is coaxial with the axis of rotation **60** of the associated second arm **72**. This allows the first and second arms **71** and **72** to be efficiently positioned on the base frame **48**. Alternatively, the axis of rotation **60** of each first arm **71** and the axis of rotation **60** of the associated second arm **72** may not be coaxial.

(146) (Drive Units)

(147) The pair of first arms **71** are driven by a first drive unit. The first drive unit rotates the pair of first arms **71** relative to the base frame **48** to change the distance between the first claws **71b** in a lateral direction (x-direction). In the implementation shown in FIGS. **14A** and **14B**, the first drive unit is constituted by actuators **82** provided for the respective first arms **71**.

(148) An actuator **82** may be an air cylinder, for example. The actuator **82** adjusts the amount of extension of an actuation shaft **82A** that moves in the axial direction. A pin **82B** is provided on the distal end of the actuation shaft **82A**. The pin **82B** is inserted into a slot in a link **90** fixed to the associated first arm **71** so as to be movable and rotatable.

(149) When each actuator **82** extends its actuation shaft **82A**, the associated first arm **71** extends downwardly, as shown in FIG. **14A**, such that the pair of first claws **71b** of the pair of first arms **71** are located closer to each other, i.e., in a closed state. When each actuator **82** retracts its actuation shaft **82A**, the pair of first claws **71b** of the pair of first arms **71** move away from each other, as shown in FIG. **14B**, such that they are in an open state.

(150) The pair of second arms **72** are driven by a second drive unit (not shown). The second drive unit rotates the pair of second arms **72** relative to the base frame **48** to change the distance between the second claws **72b** in a lateral direction (x-direction). The second drive unit that drives the second arms **72** may be an arrangement including actuators similar to the actuators **82** shown in FIGS. **14A** and **14B**, for example. The pair of second arms **72** are controlled by the second drive unit to be, again, in a closed state, in which the pair of second claws **72b** are located closer to each other, or in an open state, in which the pair of second claws **72b** are more distant from each other than in the closed state. It will be understood that the actuators of the first and second drive units are not limited to air cylinders, and may be motors or hydraulic cylinders, for example.

(151) The pair of first claws **71b** are capable of supporting the lower surface of the first workpiece **W1** at both ends as determined along the lateral direction when they are laterally closer to each other, i.e., in the closed state. The pair of second claws **72b** are capable of supporting the lower surface of the second workpiece **W2** at both ends as determined along the lateral direction when they are laterally closer to each other, i.e., in the closed state.

(152) [Exemplary Process for Manufacturing Press-Formed Product]

(153) An exemplary process for manufacturing a press-formed product using the hot-press

manufacturing line **10** shown in FIG. **13** will be described. The process for manufacturing a press-formed product includes a heating step, a first transportation step, a second transportation step, and a pressing step. The heating step and the first transportation step may be the same as in the manufacturing process of the hot-press manufacturing line **10** shown in FIG. **1**.

(154) (Heating Step)

(155) At the heating step, the first and second workpieces **W1** and **W2** are heated by the heating device **14** while being placed on the tray body **2**. The first and second workpieces **W1** and **W2** are heated while overlapping in the direction normal to the sheet surfaces of the first workpiece **W1**. The first workpiece **W1** is placed on the first group of struts **3a**. The second workpiece **W2** is placed on the second group of struts **3b**.

(156) (First Transportation Step)

(157) At the first transportation step, the first and second workpieces **W1** and **W2**, while being on the tray body **2**, are transported by the transportation table **16** from the heating device **14** to the lifting location for the transportation device **46**.

(158) (Second Transportation Step)

(159) The second transportation step includes: a sub-step in which the transportation device **46** simultaneously lifts the first and second workpieces **W1** and **W2**; a sub-step in which the first and second workpieces **W1** and **W2** are transported; a sub-step in which the first workpiece **W1** is lowered to a press location; and a sub-step in which the second workpiece **W2** is lowered to a press location.

(160) As the manipulator **44** operates, the transportation device **46** moves toward the first and second workpieces **W1** and **W2** on the tray **1** from above. At this moment, the transportation device **46** has the pair of first arms **71** and the pair of second arms **72** in the open state. The transportation device **46** moves toward the second workpiece **W2** from above, with the second workpiece **W2** on the second group of struts **3b** located between the second arms **72** as viewed from above. When the transportation device **46** has moved close enough to allow the second claws **72b** of the second arms **72** to “dive” to below the second workpieces **W2** placed on the second group of struts **3b**, the device rotates the pair of second arms **72** into the closed state. This allows the second claws **72b** of the pair of second arms **72** to dive to below the lower surface of the second workpiece **W2** at both ends. The transportation device **46** is now supporting the lower surface of the second workpiece **W2** using the second claws **72b** of the second arms **72**. FIGS. **13** and **14B** show that the transportation device **46** are supporting the second workpiece **W2** using the second arms **72**.

(161) The transportation device **46**, while supporting the second workpiece **W2** with the second arms **72**, rotates the first arms **71** into the closed state. This allows the first claws **71b** of the pair of first arms **71** to dive to below the lower surface of first workpiece **W1** at both ends. The transportation device **46** is now supporting the lower surface of the first workpiece **W1** using the first claws **71b** of the first arms **71**. In this state, the transportation device **46** moves upward as the manipulator **44** operates. Thus, the first and second workpieces **W1** and **W2** are lifted simultaneously. Although in this implementation the second arms **72**'s operation of rotating and supporting the first workpiece **W2** and the first arms **71**'s operation of rotating and supporting the first workpiece **W1** occur successively, these operations may occur simultaneously, or may occur in the reversed order.

(162) The transportation device **46** transports the workpieces while the first workpiece **W1** with its lower surface supported by the claws of the pair of first arms **71** and the second workpiece **W2** with its lower surface supported by the claws of the pair of second arms **72** overlap each other in the direction normal to the sheet surfaces of the first workpiece **W1**.

(163) FIG. **13** shows a situation immediately after the transportation device **46** has placed the first workpiece **W1** on its press location. To place the first workpiece **W1**, the transportation device **46**, while supporting the first workpiece **W1** with the claws of the pair of first arms **71** (i.e., first claws **71b**), places it on the press location for the first workpiece **W1** and, at this location, rotates the first

arms **71** into the open state. When placing the first workpiece **W1**, the transportation device **46** is continuing to hold the second workpiece **W2** with the second arms **72**. Thereafter, the transportation device **46** moves to the press location for the second workpiece **W2**, drives the pair of second arms **72** into the open state, and places the second workpiece **W2** on its press location. (164) In the implementation shown in FIG. **13**, two workpieces are positioned between the upper and lower die parts **21** and **23** of a single press machine and the two workpieces are press-formed simultaneously. In this implementation, the first and second workpieces **W1** and **W2** are lowered onto different press locations between the upper and lower die parts **21** and **23**. The lowering of the two workpieces held by the transportation device **46** onto their respective press locations is not limited to this manner. For example, the press locations may be provided in two different press machines, where the first and second workpieces **W1** and **W2** are lowered onto their respective press locations.

(165) [Experiment]

(166) A steel sheet was heated and temperature changes after completion of heating were measured under various conditions. Specifically, the experiment was conducted as follows: The samples used were steel sheets for hot pressing of the order of 1.5 GPa having a sheet thickness of 0.8 mm or 1.6 mm, and temperature was measured by thermocouples attached to a surface of the steel sheet. The steel sheet was heated in a heating furnace to 950° C., and temperature drop after removal from the heating furnace and during air cooling was measured. For Comparative Example 1, temperature was measured where a single steel sheet with a sheet thickness of 1.6 mm was heated alone and left to cool; for Comparative Example 2, a single steel sheet with a sheet thickness of 0.8 mm was heated alone and left to cool; for the inventive examples, two steel sheets with a sheet thickness of 0.8 mm were positioned to overlap each other in the direction normal to the sheet surfaces and fixedly arranged with a predetermined distance **D**. Three values of the distance **D** between the two overlapping steel sheets were used: 10 mm, 30 mm, and 50 mm, which provided Inventive Example 1, Inventive Example 2, and Inventive Example 3, respectively. FIG. **15** indicates locations at which temperature was measured. The temperature of each of the upper and lower steel sheets was measured at positions of 5 mm, 20 mm, 30 mm, and 50 mm away from an edge of the steel sheet, and at the center of the steel sheet.

(167) FIG. **16** is a graph of measurements of average temperature drop rate. The average temperature drop rate was derived across the section for 800° C. to 750° C. along the temperature drop curve from measurements. FIG. **17** is a graph indicating the range from which the average temperature drop rates were derived. The results shown in FIG. **16** demonstrate that, for all the measurement locations, the average temperature drop rates for Inventive Examples 1 to 3, which had two steel sheets overlapping in the top-bottom direction with a sheet thickness of 0.8 mm, were lower than that for Comparative Example 1, which had a single steel sheet with a sheet thickness of 0.8 mm. The results also demonstrate that, with two overlapping steel sheets, examples with lower values of the distance **D** between the two steel sheets achieved lower average temperature drop rates near the edges of the steel sheets and achieved greater improvements in average temperature drop rate for all the measurement locations. Experiment 3, which had a distance **D** of 50 mm, improved the temperature drop rate to an extent approximately in the middle between that for Comparative Example 1, which had a single steel sheet with a sheet thickness of 0.8 mm, and that for Comparative Example 2, which had a single steel sheet with a sheet thickness of 1.6 mm. Inventive Example 1, which had a distance **D** of 10 mm, improved the temperature drop rate to an extent substantially equal to that of Comparative Example 2, which had a single steel sheet with a sheet thickness of 1.6 mm. Setting an appropriate distance **D** between the two overlapping steel sheets achieved temperature-drop characteristics substantially equal to those of a steel sheet with a sheet thickness that was twice as large, i.e., with a heat capacity that was twice as high.

(168) These results show that, when a plurality of workpieces are transported while overlapping in the top-bottom direction, keeping the workpieces at an appropriate distance is important to reduce

temperature drop rate. The above-described embodiments enable retaining the distance between the first and second workpieces placed on the tray in a stable manner during the heating step and the subsequent transportation step. This will reduce temperature drop in the first and second workpieces and, at the same time, enable simple and efficient transportation.

REFERENCE SIGNS LIST

(169) **1**: tray **2**: tray body **3**: struts **3a**: first group of struts **3b**: second group of struts **W1**: first workpiece **W2**: second workpiece

Claims

1. A method of manufacturing a press-formed product comprising: a heating step in which a sheet-shaped first workpiece and a sheet-shaped second workpiece are simultaneously heated; a transportation step in which the first workpiece and the second workpiece heated at the heating step are transported to a press machine; and a pressing step in which the first workpiece and the second workpiece transported to the press machine are processed by the press machine, wherein, at the heating step, the first workpiece and the second workpiece are heated by a heating device while the first workpiece is placed on a first group of at least three struts extending upward from a tray body having an empty space extending therethrough in a top-bottom direction when viewed from above and the second workpiece is placed on a second group of at least three struts extending upward from the tray body and positioned above the first workpiece to overlap the first workpiece in a direction normal to a sheet surface of the first workpiece, wherein the transportation step includes: a first transportation sub-step in which the first workpiece and the second workpiece, together with the tray body, are transported from the heating device to a lifting location while the first workpiece is placed on the first group of struts and the second workpiece is placed on the second group of struts and positioned above the first workpiece to overlap the first workpiece in the direction normal to the sheet surface of the first workpiece; and a second transportation sub-step in which the second workpiece placed on the second group of struts, when at the lifting location, is lifted upward by a transportation device and transported to a press location for the second workpiece, and the first workpiece placed on the first group of struts, when at the lifting location, is lifted upward by the transportation device and transported to a press location for the first workpiece.

2. The method of manufacturing the press-formed product according to claim 1, wherein, at the heating step and the first transportation sub-step, the first workpiece placed on the first group of struts and the second workpiece placed on the second group of struts are positioned to overlap in the top-bottom direction so as to be separated by a distance not larger than 100 mm.

3. The method of manufacturing the press-formed product according to claim 1, wherein, at the heating step and the first transportation sub-step, a relationship between a maximum distance D (mm) between the first workpiece placed on the first group of struts and the second workpiece placed on the second group of struts as measured in the top-bottom direction and a minimum sheet thickness t (mm) of a thinnest portion of the first workpiece and the second workpiece is represented by an expression:

$$D \leq 120t.$$

4. The method of manufacturing the press-formed product according to claim 1, wherein: an area of a sheet surface of the second workpiece is larger than an area of the sheet surface of the first workpiece; and, at the heating step and the first transportation sub-step, the first workpiece is entirely placed on the first group of struts and the second workpiece placed on the second group of struts overlap in the top-bottom direction.

5. The method of manufacturing the press-formed product according to claim 4, wherein, at the heating step and the first transportation sub-step, a minimum value of a distance ΔW between an edge of the second workpiece placed on the second group of struts as viewed from above and an edge of the first workpiece placed on the first group of struts as viewed from above is not smaller

than 5 mm.

6. The method of manufacturing the press-formed product according to claim 4, wherein, at the heating step and the first transportation sub-step, a maximum value of a distance ΔW between an edge of the second workpiece placed on the second group of struts as viewed from above and an edge of the first workpiece placed on the first group of struts as viewed from above is not larger than 30 mm.

7. The method of manufacturing the press-formed product according to claim 1, wherein a sheet thickness of the second workpiece is larger than a sheet thickness of the first workpiece.

8. The method of manufacturing the press-formed product according to claim 1, wherein, at the first transportation sub-step, the first workpiece placed on the first group of struts and the second workpiece placed on the second group of struts are transported while being covered with a shield from a front as determined along a direction of transportation.

9. The method of manufacturing the press-formed product according to claim 8, wherein the shield includes an inclined surface inclined from a center of the shield toward an end of the shield so as to be located closer to the first workpiece and the second.

10. The method of manufacturing the press-formed product according to claim 8, wherein: at least one of the first workpiece placed on the first group of struts and the second workpiece placed on the second group of struts has a long-length direction and a short-length direction; and, at the first transportation sub-step, the shield covers the first workpiece placed on the first group of struts and the second workpiece placed on the second group of struts along the long-length direction.

11. The method of manufacturing the press-formed product according to claim 1, wherein: each of the first workpiece and the second workpiece is a differential-thickness sheet with a large-thickness portion and a small-thickness portion; and, at the heating step and the first transportation sub-step, the large-thickness portion of the first workpiece placed on the first group of struts overlaps, as viewed in the top-bottom direction, the small-thickness portion of the second workpiece placed on the second group of struts.
